

Timothy Asady-Kany

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EDUCATION

University of Central Florida | B.S. Aerospace Engineering

Expected: December 2028

GPA: 3.368/4.00 | Completed Credit Hours: 80

ENGINEERING EXPERIENCE

Fluid Responsible Engineer

Orlando, FL | *January 2026 - Present*

Knights Experimental Rocketry (KXR)

Hours/week: 10

- Collaborated with multidisciplinary engineering teams to develop and integrate a fluid system for a liquid rocket engine test stand, and eventually a flight vehicle.
- Supported fluid system design through P&ID development, selecting components while balancing cost and performance, and presenting system-level designs through preliminary and critical design reviews.
- Developed MATLAB scripts to calculate mass flow rates, pressure requirements, and component sizing, using results to verify system requirements and select the correct components prior to design reviews.
- Modeled fluid-system components and routing in SolidWorks, integrating manifolds, tubing, valves, and other hardware into the overall test-stand CAD model.
- Developed technical documentation including system test and operational procedures, system requirements, and Concept of Operations to ensure safety while operating the high-pressure system.
- Led fluid system integration after our team's original test stand's cancellation due to budget constraints; redesigned the plumbing system to fit onto an existing test stand while keeping original system requirements.
- Conducted component test procedures for ball valves, tubing, piping, and mounting hardware for WANDA, a mobile k-bottle ground-support cart designed to easily interface with the engine test stand.

PROJECTS

Self-Balancing Robot

- Designed and built a self-balancing BDX-inspired droid using SolidWorks, ESP32, and 3D printing, wiring electrical components together and performing full system tests as well as individual hardware tests.
- Implemented and tuned a PID control algorithm using IMU sensor data in Arduino IDE to achieve autonomous balancing and used MATLAB to accurately control the robots' motion using a Bluetooth controller.
- Created detailed SolidWorks parts, assemblies, and drawings with dimensional clearances, fastener tolerances, and internal component packaging to ensure proper fit, leading to more efficient manufacturability.

L1 High-Powered Rocket

- Modeled and simulated a Level 1 high-power rocket in OpenRocket, evaluating multiple motor configurations, fin geometries, and nose cone designs to optimize flight performance with a stability of ~ 1.8 cal and predicted apogee of 574m.
- Designed the rocket in SolidWorks, using CAD to determine accurate dimensions and mass properties for fabrication.

Wind Tunnel

- Designed a small-scale wind tunnel in SolidWorks to visualize aerodynamic flow and test scaled cars, rockets, and aircraft models.
- Designed the test section, honeycomb flow straightener, 6:1 contraction cone, and diffuser to improve flow uniformity and reduce turbulence while focusing on manufacturable design choices using a 3D printer.

SKILLS AND COURSEWORK

- **Experience with:** SolidWorks, Fusion 360, OnShape, OpenRocket, Microsoft Office (Excel, Word, PowerPoint, Outlook), GitHub
- **Programming skills:** C, Python, MATLAB, HTML, CSS, Arduino IDE
- **Relevant Coursework:** Properties of Aerospace Materials, Mechanical Engineering Problem Solving, Statics, Principles of Electrical Engineering
- **Languages:** English (Native), French (Fluent)